

El Moustpha Cheikh JIDDOU

Engineering Student at the Higher Polytechnic School

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PROFESSIONAL EXPERIENCE

Data Systems & IT Architecture Intern – Banque Nationale de Mauritanie (BNM) February 2025 – Present

 Nouakchott, Mauritania

- Designed and implemented a centralized enterprise Data Warehouse backed by robust ELT pipelines, automated flow orchestration, and predictive models.
- Architected and deployed a containerized microservices infrastructure over isolated Docker networks on on-premise production servers.
- Built a secure conversational agent (Click) connected to the WhatsApp REST API, including a custom NLU module for dynamic query routing.
- Implemented a RAG (Retrieval-Augmented Generation) architecture with a PostgreSQL/pgvector database to index and query official banking documentation without data leakage.
- Benchmarked several large language models (GPT-4o, Claude 3.5, Gemini 1.5, local Llama) to optimize the trade-off between performance, infrastructure cost, and strict confidentiality.
- Developed sequential neural network models (LSTM, GRU) to anticipate rapid photovoltaic power drops and plan for critical infrastructure resilience.
- Technical environment:** Python (FastAPI), Data Warehousing, ELT pipelines, Docker, N8N, PostgreSQL (pgvector), PyTorch, React, REST APIs, Linux/Bash.

End-of-Studies Internship – Macroeconomic Analysis and DSGE Modeling – Banque Centrale de Mauritanie (BCM) – Directorate General of Economic Studies and Monetary Stability February 2026 – May 2026

 Nouakchott, Mauritania

- Produced a master's thesis in Econometrics and Applied Statistics on the comparative analysis of monetary regimes in Mauritania using a small open-economy DSGE model.
- Analyzed monetary policy transmission mechanisms, inflation, exchange rate dynamics, and macroeconomic dynamics in the Mauritanian context.
- Structured and exploited macroeconomic and monetary time series from BCM, ANSADE, and international databases for calibration and economic simulations.
- Participated in the activities of the Directorate General of Economic Studies and Monetary Stability: conjunctural analysis, macroeconomic forecasts, monetary statistics, and economic studies.
- Developed economic modeling and simulation tools to compare several monetary policy regimes and assess their impact on macroeconomic stability.
- Produced quantitative analyses, economic charts, and statistical indicators to support results interpretation and decision-making.
- Environment:** Econometrics, DSGE, macroeconomic analysis, monetary statistics, Python, R, Excel, Julia, GAMS, economic modeling, data analysis.

Lecturer – Poverty & Household Living Conditions + Sociological Survey Methods – Institut Supérieur des Métiers de la Statistique (ISS) February 2026 – June 2026

 Nouakchott, Mauritania

- Renewed for a second consecutive year with an additional course module.
- Taught the Poverty and Household Living Conditions module (improved edition).
- Designed and delivered a new module: Sociological Survey Methods (survey methodology, sampling, questionnaire design, data analysis).
- Supervised tutorials and practical sessions with field survey simulations using KoboToolbox, CSPro, and Stata.
- Directed and evaluated final-year projects (PFE) and supervised worker internships.
- Developed course materials, case studies, and evaluation frameworks for both modules.

Lecturer – Poverty and Household Living Conditions – Institut Supérieur des Métiers de la Statistique (ISS) February 2025 – June 2025

 Nouakchott, Mauritania

- First higher education teaching experience at the institution from which I graduated.
- Prepared and delivered lectures on poverty measurement methodologies and household living conditions analysis.
- Designed and supervised tutorials (TD) and practical sessions (TP), using real case studies from national household surveys.
- Provided personalized academic support and continuous student monitoring throughout the semester.
- Supervised final-year projects (PFE) and worker internships, with mentoring on methodological and technical aspects.

Assistant Engineer Intern – Mining Department, Geotechnical Service – Kinross Gold Corporation (Tasiast Mauritania) July 2025 – September 2025

 Tasiast, Mauritania

- Wall Compliance & 80/80:** designed a Python/Gradio mini-dashboard for *Toe/Crest/Berm Width/BFA* monitoring from CALC and COMPLIANCE files (auto-detection of inputs, KPIs, 80/80 bars, trends & deltas).
- Daily Survey digitization:** Microsoft Forms → SharePoint/Excel Web → Power BI pipeline with typing/validation, duration calculations, KPIs (% Done, Backlog, SLA Breach), and near real-time refresh.

- **Short-term PV forecasting:** preparation of minute-level data (*lags*, ramps ΔP , time encodings) and sequential prototypes (LSTM/GRU) to anticipate rapid power drops and improve alert *lead time*.
- **Practices:** git versioning, technical documentation, skills transfer to field/process teams.
- **Environment:** Python (NumPy, pandas, Plotly, Gradio), Power BI, SharePoint/Excel Web.

President & Event Coordinator – ESP Data Club

📅 *October 2022 – April 2025*

📍 *Nouakchott, Mauritania*

- **Event Coordinator** (October 2022 – January 2023): Organized events to promote statistics and data analysis. Coordinated with members to ensure the success of activities.
- **President** (January 2023 – April 2025): Led the club, set strategic objectives, managed data collection and analysis projects, developed academic partnerships.
- Skills: Leadership, project management, event planning, communication.
- See the Club.

End-of-Study Internship – Caisse des Dépôts et de Développement (CDD)

📅 *February 2023 – June 2023*

📍 *Nouakchott, Mauritania*

- Analyzed banking data from CDD to assess the impact of the credit guarantee system on loan defaults in Mauritania.
- Used econometric models and modern machine learning approaches to predict default risks.
- Prepared and interpreted statistical reports on the socio-demographic and financial characteristics of borrowers.
- Developed computational solutions for risk analysis using algorithms such as logistic regression, k-NN, and Random Forest.
- See the analyses.

Enumerator – Agence Nationale de la Statistique et de l'Analyse Démographique et Économique (ANSADE)

📅 *July 2022 – August 2022*

📍 *Nouakchott, Mauritania*

- Collected field data for the pilot General Population and Housing Census (RGPH).
- Skills acquired: data collection, communication, fieldwork.

Worker Internship – Agence Nationale de la Statistique et de l'Analyse Démographique et Économique (ANSADE)

📅 *April 2022 – June 2022*

📍 *Nouakchott, Mauritania*

- Developed a methodology for data mapping to design sectoral accounts.
- Participated in the collection and processing of economic branch data for national accounts.
- Analyzed sectoral data using ERETES software to compile economic accounts.
- Trained on GDP decomposition techniques as part of the International Comparison Program.
- Contributed to the national survey process for the mobilization and quantification of ZAKAT, including enumerator training.

PROJECTS

Grain Size Analysis of 316L Microstructure – Academic project: Computer-vision grain size analysis

📅 *June 2025 – August 2025*

📍 *Nouakchott, Mauritania*

- Developed a complete image-processing pipeline to segment grain boundaries in 316L stainless steel from 500× micrographs.
- Implemented and compared segmentation methods: classical thresholding, gradient filters, the HED model (Caffe/OpenCV), and a convolutional U-Net trained on real and synthetic data.
- Automatic extraction of grain size in physical units (μm , μm^2) via pixel/micron calibration and computation of statistics (size distribution, axes, circularity).
- Created an interactive interface (Gradio) to load a micrograph, segment grains, and export geometric features to CSV.
- See the GitHub repository.

RAG-WebSystem – Academic project: Generative AI system based on Retrieval-Augmented Generation

📅 *March 2025 – April 2025*

📍 *Nouakchott, Mauritania*

- Designed a RAG engine (Retrieval + LLM) able to answer queries by drawing on external documents indexed by vectors.
- Implemented a complete pipeline: ingestion, vectorization (FAISS), semantic search, then contextual answer generation with an open-source model.
- Built a REST API (FastAPI) and a reactive web interface (React + Tailwind) to test, trace, and evaluate the answers.
- See the GitHub repository.

DataMine App – Academic project: Corporate industrial project

📅 *November 2023 – Present*

📍 *Nouakchott, Mauritania*

- Designed and developed a collaborative platform for managing and tracking tasks on a geostatistical data project.
- Analyzed geostatistical data to optimize resource management and improve decision-making.
- Created an intuitive user interface for managing and visualizing tasks on GitHub.
- See the project here.

Agricultural Data Analysis System (Smart Greenhouse) – Academic project: Entrepreneurship and innovation project

📅 *November 2023 – June*

2024

📍 Nouakchott, Mauritania

- Designed and developed an intelligent system for analyzing agricultural data, specifically for greenhouses.
- Implemented IoT sensors for real-time data collection and created interactive dashboards for monitoring.
- Applied machine learning techniques to optimize resource management and improve agricultural yields.
- See the project [here](#).

Disaster Tweets NLP Classification – Academic project: Introduction to big data 📅 January 2022 – February 2022

📍 Nouakchott, Mauritania

- Developed a machine learning model to classify tweets related to natural disasters.
- Processed massive data and applied advanced NLP (Natural Language Processing) techniques.
- Used Python libraries to build and evaluate the model.
- See the project [here](#).

Survey Management Application – Academic project: Advanced databases 📅 December 2022 – January 2022

📍 Nouakchott, Mauritania

- Designed and built a survey management application to facilitate data collection and tracking.
- Backend development with the Django framework and front-end creation in HTML, CSS, and JavaScript.
- See the project [here](#).

Survey on the Perception of Influencers – Academic project: Sociological surveys 📅 January 2022 – March 2022

📍 Nouakchott, Mauritania

- Designed and conducted a survey on the perception of influencers in Mauritanian society.
- Developed the questionnaire on KoboCollect to ensure structured data collection.
- Independently collected and analyzed the data, interpreting results to identify social trends.

Grade Management Application – Academic project: Advanced databases and algorithms 📅 December 2022 – January 2022

📍 Nouakchott, Mauritania

- Designed and built a grade management application for students.
- Backend development with the Flask framework and use of HTML, CSS, and JavaScript for the front-end.
- See the project [here](#).

Determinants of Academic Performance in Mauritania – Academic project: Statistical software 📅 May 2021 – June 2021

📍 Nouakchott, Mauritania

- Analyzed the factors influencing academic performance in Mauritania using R.
- Studied data from CONFEMEN's education systems (PASEC1), including socio-demographic, geographic, and pedagogical variables.

SOFT SKILLS

Initiative • Curiosity • Interpersonal Ease • Creative • Listening Skills

EDUCATION

Higher Polytechnic School of Nouakchott – ESP - Mauritania 📅 October 2023 – June 2026
Degree: State Engineer in Statistics and Data Engineering.

National School of Computer Science and Systems Analysis – ENSIAS - Rabat 📅 October 2025 – February 2026
Exchange semester: Computer Engineering and Networks.

Faculty of Economic Sciences and Management – FSEG - Mauritania 📅 October 2023 – June 2025
Degree: Master's in Applied Econometrics and Statistics – Economic Statistical Engineering Track.

Polytechnic Group – Higher Institute of Statistical Professions – GP - Mauritania 📅 October 2020 – June 2023
Degree: Professional Bachelor's in Applied Statistics.

Atar High School – Atar – Mauritania 📅 2019 – 2020
Degree: Baccalaureate, Mathematics track (Bac C).

SKILLS

- **Analysis & Modeling:** statistics, econometrics, surveys; data mining (classification, regression, clustering, association rules); time series (short-term forecasting).
- **Big Data, Data Engineering, DB & GIS:** Architecture & modeling of a centralized Data Warehouse, ELT/ETL pipelines (Airbyte, dbt), data flow orchestration (N8N), Hadoop, Spark, Kafka, MapReduce, Hive SQL; SQL management & optimization (execution plans, indexing, partitioning, Oracle SQL Tuning); Docker & Dev Containers; Stata, SPSS, Excel, GAMS, Eviews, Sphinx, KoboToolbox, ODKCollect, CSPro; Oracle SQL Developer; PostgreSQL/PostGIS (incl. pgvector); QGIS/ArcGIS; GIS integration & analysis (spatial joins, basic geoprocessing).
- **AI, MLOps & Programming:** supervised ML (regression, SVM, random forests, decision trees); deep learning (LSTM/GRU for time series, CNN, Transformers) with TensorFlow, PyTorch, Keras; NLP & RAG (embeddings, language models, augmented retrieval, RAGAI API); MLOps (MLflow, Docker/Compose); Python/PySpark, Julia, R, Scala, SQL (incl. HiveQL), PL/SQL, Java (J2EE, **Spring Boot, Spring Cloud, Spring Data, Spring Security, Spring microservices**), VBA, $\text{\LaTeX}$; Bash & Linux CLI; Git/GitHub; Jira, Trello, VS Code, Cursor; web development (HTML, CSS, JavaScript, Django, Flask, REST APIs, J2EE middleware integration), React, Flutter.

LANGUAGES

Arabic Native Language	
French DALF C1	
English CEFR B1 - Tracktest English Assessment	

VOLUNTEERING

- President, ISS Alumni Association (current position)
- Events & Programs Officer, Association des Statisticiens Mauritaniens
- President, ESP Data Club (January 2023 – April 2025)
- Event Coordinator, ESP Data Club (October 2022 – January 2023)
- Member, Clubs Leaders Étudiants Francophones (CLEF)
- Member, Mauritanian Engineering Community (MEC)
- Military Training

ACHIEVEMENTS

 1st Place, RIMAI Data Science Challenge 2024 Competition focused on license plate recognition using AI techniques. Learn more about the challenge at the RIMAI Challenge Website.
 Organizer, First Datathon in Mauritania Three-day competition centered on Mauritanian data analysis. As president of the ESP Data Club, I supervised the organization and judged competitors through competitive data science challenges.

CERTIFICATIONS

freeCodeCamp (6): Relational Database (v8) (2025) • Data Visualization (2023) • Data Analysis with Python (2022) • Scientific Computing with Python (2022) • Front End Development Libraries (2022) • Responsive Web Design (2022)

DataCamp (9): Data Manipulation in SQL (2023) • Intermediate SQL (2023) • Introduction to Data Science in Python (2023) • Python Data Science Toolbox (Part 2) (2023) • Understanding Data Engineering (2023) • Writing Efficient R Code (2023) • Intermediate Python (2022) • Intermediate R (2022) • Introduction to Version Control with Git (2022)

Google (3): Google Advanced Data Analytics Capstone (2023) • Google Advanced Data Analytics Professional Certificate (2023) • Google Data Analytics (2023)

INTERESTS

- Coding
- Travel
- Football
- Tourism